INCH-POUND
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# COMMERCIAL ITEM DESCRIPTION

## FURNACES AND OVENS, HEAT-TREATING, ELECTRICALLY HEATED NATURAL OR INERT ATMOSPHERE, BATCH/BOX TYPE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers electrically heated batch/box type furnaces and ovens intended for use in heat-treating processes including solution heat treating, aging, annealing, stress relieving, precipitation hardening, hardening, tempering, and normalizing metal parts in a natural or inert atmosphere for use in Aircraft Metals Technology Shops across the Air Force. Furnaces covered by this CID are not intended for controlled atmosphere use other than inert gas (argon or nitrogen) for certain high temperature applications to minimize oxidation.

1.1 Intended use. A furnace and oven are interchangeable terms used when talking about heat treating equipment. Generally, furnaces operate at higher temperatures with radiant heat transfer, and ovens operate at lower temperatures with convection heat transfer. These furnaces and ovens consist of an insulated chamber with an external reinforced steel shell, a heating system for the chamber and one or more access doors. The types of furnaces and ovens covered herein shall be electrically heated box/batch design, used for "heat-hold-cool" temperature cycles with a water or oil quench when required. That is, the furnace or oven shall be designed to be repeatedly heated from room temperature to a maximum temperature, at a controlled rate, held at a temperature, and then cooled at a controlled rate or as required by the aerospace material specification for the particular alloy being heat treated. Heat treating cycles shall be executed by a programmable controller and be capable of being operated by Air Force Specialty Code 2A7X1. A data acquisition system shall be provided that readily produces a secure and irrefutable record of the heat treating cycle and includes all related batch documentation.

2. CLASSIFICATION. The furnaces and ovens shall be of the following types, styles, ranges, and sizes.

2.1 Type. Type designates the number of chambers a furnace or oven has (one or two):

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Type I – Single chamber.

Type II – Double (dual) chamber.

2.2 Style. Style refers to temperature ranges and heat transfer:

2.2.1 Style A - Forced air convection design:

Class 1 – 500 °F maximum operating temperature.
Class 2 – 1200-1600 °F maximum operating temperature.

2.2.2 Style B - Radiant heat design:

Class 3 – 1200 °F maximum operating temperature
Class 4 – 1400 °F maximum operating temperature
Class 5 – 2000-2500 °F maximum operating temperature

2.3 Range. Range refers to furnaces and ovens maximum and minimum operating temperatures.

2.3.1 Range 1. 500 °F maximum operating temperature; 200 °F minimum operating temperature. At a minimum, a furnace or oven shall meet ±10 °F uniformity.

2.3.2 Range 2. 1200-1600 °F maximum operating temperature; 200 °F minimum operating temperature. At a minimum, a furnace or oven shall meet ±10 °F uniformity. Style B shall be capable of inert atmosphere use with gas-tight design and door seals.

2.3.3 Range 3. 2000-2500 °F maximum operating temperature; at least 1200 °F minimum operating temperature (can be lower than 1200 °F). At a minimum, a furnace or oven shall meet ± 25 °F uniformity. Style B shall be capable of inert atmosphere use with gas-tight design and door seals.

2.4 Size and depth. Standard work volume width, height, and depth are shown in tables I through III. The work volume height, width and depth shall only refer to the guaranteed/certified Temperature Uniformity Survey (TUS) work volume where parts can be placed. This is not the internal surface to surface (wall-to-wall) dimensions of the chamber. The hearth (3.6.3) of a furnace is where parts shall be placed and would always have the dimensions of the width and depth of the work volume and could also mean the lowest setting height of a basket or rack. The hearth, therefore, as defined herein, is not the furnace “floor”, but an elevated surface for parts placement. This area shall be clearly identified by the contractor via associated literature and/or markings within or on the furnace/oven.

TABLE I. Sizes of type I, style A, range 1 or 2 furnaces and ovens.

| Size | Work Volume Dimensions, Inches Dimensions, Inches | | | Max Temperature °F / Range |
|---|---|---|---|---|
| | Width | Height | Depth | |
| 01 | 18 | 12 | 36 | 500 / 1* |
| 02 | 24 | 16 | 48 | 500 / 1* |
| 03 | 12 | 12 | 36 | 1200 / 2 |
| 04 | 24 | 16 | 48 | 1200 / 2 |
| 05 | 18 | 12 | 35 | 1200 / 2 |
| 06 | 24 | 24 | 36 | 1200 / 2 |
| 07 | 30 | 16 | 72 | 1200 / 2 |

NOTE: All range 2 furnaces and ovens shall include a stainless-steel quench tank for water.
*Quench tank not included for Range 1, style A (500 °F) ovens.



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TABLE II. Sizes of type I, style B, range 2 or 3 furnaces and ovens.

| Size | Work Volume Dimensions, Inches | | | Max Temperature °F / Range |
|---|---|---|---|---|
| | Width | Height | Depth | |
| 08 | 12 | 15 | 30 | 1200 / 2 |
| 09 | 18 | 24 | 36 | 1400 / 2 |
| 10 | 12 | 12 | 24 | 2000 / 3 |
| 11 | 18 | 12 | 48 | 2000 / 3 |
| 12 | 24 | 36 | 72 | 2000 / 3 |
| 13 | 18 | 24 | 36 | 2500 / 3 |

NOTE: All range 3 furnaces and ovens shall include a stainless-steel quench tank for oil.

TABLE III. Sizes, style, and range of type II furnaces and ovens.

| Size | Style | Range | Work Volume Dimensions, Inches | | | Quench Tank | Max Temperature °F (Upper / Lower) |
|---|---|---|---|---|---|---|---|
| | | | Width | Height | Depth | | |
| 14 | A<br>A | 1<br>2 | 24<br>24 | 16<br>16 | 48<br>48 | <br>Water | 500<br>1200 |
| 15 | A<br>A | 2<br>1 | 30<br>30 | 16<br>16 | 72<br>72 | Water | 1200<br>500 |
| 16 | A<br>A | 2<br>1 | 12<br>12 | 12<br>12 | 36<br>36 | Water | 1200<br>500 |
| 17 | A<br>A | 2<br>2 | 24<br>24 | 24<br>24 | 36<br>36 | Water | 1200<br>1200 |
| 18 | B<br>A | 2<br>3 | 18<br>12 | 12<br>12 | 48<br>36 | Water<br>Oil | 1200<br>2000 |
| 19 | B<br>A | 2<br>3 | 24<br>18 | 16<br>12 | 48<br>48 | Water<br>Oil | 1200<br>2000 |
| 20 | B<br>A | 3<br>2 | 12<br>12 | 12<br>12 | 36<br>36 | Oil<br>Water | 2000<br>1200 |
| 21 | B<br>A | 3<br>2 | 24<br>24 | 24<br>24 | 36<br>36 | Oil<br>Water | 2000<br>1200 |
| 22 | B<br>A | 3<br>2 | 18<br>18 | 24<br>24 | 48<br>48 | Oil<br>Water | 2300<br>1600 |

## 3. SALIENT CHARACTERISTICS.

3.1 General requirements. The furnace or oven shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements specified herein. The furnace shall be an electrically heated box/batch design having electrical heating elements, a hearth or with a base tray, rack, or basket (to ensure that parts are not placed onto the furnace floor) to support the load, and a single opening in the end of the chamber for loading and unloading of parts on a horizontal plane.

3.2 Design requirements. The furnace or oven shall be designed to be repeatedly heated from room temperature to a maximum designated temperature, at a controlled rate, held at a temperature, and then cooled at a controlled rate as required by the associated aerospace material specification (example Society of Automotive Engineers/Aerospace Materials Specifications (SAE AMS) 2750).



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Heat treating cycles shall be executed by a programmable controller and be easy to use by an operator (example: Air Force Specialty Code 2A7X1).

3.3 Construction materials. The furnace or oven shall be constructed of parts that are new, without defect(s), and free of repairs or refurbishment. The structure shall be capable of repeatedly withstanding temperature variations encountered during operation of the furnace or oven to its maximum rating and capacity without permanent distortion.

3.4 Components. Furnaces and ovens shall include all components, parts, and features necessary to fully meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.4.1 Edges. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.4.2 External finish. The external finish shall be industrial enamel or water based heavy duty paint, properly applied and suitable for the environment intended. The color shall be the manufacturer's standard color scheme, or a color agreed upon by the Air Force Metals Technology Office / Metal Fabrication and Technologies Office.

3.4.3 Heating power source. Shutdown of the heating power source shall not inadvertently affect the operation of equipment such as ventilation or recirculating fans, cooling components, safety and/or other auxiliary equipment.

3.5 Furnaces and ovens.

3.5.1 Furnace type:

3.5.1.1 Type I. Type I furnaces and ovens shall have a single chamber of either a style A forced air convection design or a style B radiant heat design.

3.5.1.2 Type II. Type II furnaces and ovens shall have a double (dual) chamber (over/under) with a style A lower chamber and a style B upper chamber or, a style A upper and lower chamber.

3.5.2 Furnace style:

3.5.2.1 Style A. Style A furnaces and ovens shall use forced air convection to heat the load. Air shall be heated in an area separated from the load and blown uniformly into the chamber by one or more fans. The load shall not be exposed to direct radiant heat from the electrical heating elements and shall be protected by a physical barrier (liner/sleeve) manufactured from materials capable of withstanding temperatures greater than the maximum operating range. Style A furnaces and ovens, range 2 and operating greater than 800 °F shall have a radiation survey performed during the qualifying TUS and performed at the maximum intended operating temperature in accordance with the SAE AMS 2750 with specified panels and necessary equipment provided by the furnace supplier.

3.5.2.2 Style B. Style B furnace shall use radiant heat to heat the load by direct exposure to the electrical heating elements.

3.6 Components. All the furnaces and ovens shall include, at a minimum, the following components:

3.6.1 Frame or integrated structure. The furnace and oven shall have a structural frame that shall provide stable support for the chamber, hearth, and maximum rated load. A floor-mounted furnace or oven shall have provisions for being lifted and moved by a forklift. The



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metal frames of furnaces shall be electrically grounded. Floor mounted furnaces or ovens shall have suitable holes (ease of access/locations, sizes, numbers, etc.) for securing in place with floor lags.

3.6.2 Outer shell. The outer shell of the furnace shall be constructed of reinforced sheet steel and have welded, bolted or riveted joints to provide a rigid structure. The shell thickness shall not be less than 10 gauge.

3.6.3 Hearth. The hearth shall be constructed of heat conducting material considered by the manufacturer as standard. The hearth shall be capable of supporting repeated rated loads at the furnace or oven maximum operating temperature without permanent distortion. If required, rollers or slide rails shall be provided for moving the load in and out of the chamber. A fixture or tray to hold the load shall also be provided to protect the floor of the furnace or oven and be constructed of a material capable of repeatedly withstanding temperatures greater than the maximum operating temperature without deformation, distortions, off gassing, droop or material degradation.

3.6.4 Lining. The furnace or oven chamber and door lining shall be rated for the furnace and oven maximum operating temperature and shall have insulating properties that maintain the furnace exterior surface temperature no greater than 160 °F.

3.6.5 Door. The door of a range 1 or range 2 furnace shall be mounted on hinges and swing open. The door of a range 3 furnace shall open either by being lifted vertically either by manual means with a suitable counterweight(s), by a power-operated system, or via mounted on hinges and swing open. If a door is lifted vertically there shall be safety mechanism(s) and safety measures in place to prevent equipment damage and/or operator harm due to mechanical failure(s), human error(s), etc., in the event the primary door mechanical lift/opening system fails. As an example, a chain driven lift system shall have a mechanical fail-safe means to prevent the opened door from falling, closing, partially closing, etc., in the event chain failure (breaking), gear slipping, or similar, occurs. If the door opens via hinges, there shall be procedures and/or safety precautions to ensure the door does not swing back towards the operator and/or closure without the operator causing that to happen.

3.6.5.1 Interlock switch. The furnace and oven shall have a door-operated interlock switch that turns off the power to the heating elements and forced air convection when the door is not properly sealed and fully closed.

3.6.5.2 Door mechanism. The door mechanism shall move the door away from the face of the furnace a sufficient distance to prevent damage to the chamber perimeter seal or to the lining during lifting and lowering. For vertically lifted doors, the door hot side shall face away from the operator at all times during lifting and lowering. For doors that open via hinges (swing open), the door shall require door opening swing (sweep) of at least 180 degrees to minimize the door face material from facing and/or harming the operator. Swing doors are most ideal for either style A or B furnaces and ovens as this typically provides the fastest quench time possible when required with a one-man operation.

3.6.6 Temperature controller/recorder. The furnace shall have a programmable temperature controller and a recorder/data acquisition system/software that meets SAE AMS 2750.

3.6.6.1 Temperature controller. The temperature controller shall measure the temperature and precisely control the input of heat into the furnace or oven. The temperature controller shall maintain the respective furnace or oven chamber temperature at the control thermocouple within ±2 °F for both style A and style B furnaces and ovens of the set temperature following



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recovery to set point and stabilization. Programmable ramp and soak period programs/profiles for process temperature control shall be provided and defined by the required heat treatment process for the specific material being heat treated. The controller shall be capable of at least a minimum of 32 different process programs/profiles.

3.6.6.1.1 Control system alarm. The controller/control system shall provide an audible and visual alarm for notifying the operators of a temperature deviation condition or when the heating cycle has ended and when parts can be removed. The alarm circuit shall have a silence interlock.

3.6.6.2 Temperature recorder. The temperature recorder shall record the process control temperature via a separate thermocouple positioned within 0.38 inch of the chamber temperature control thermocouple. Parallel connection of a thermocouple to more than one instrument is forbidden. Controllers that include recording functions and computer software that are fully compliant with SAE AMS 2750 are allowed. All control, recording and high limit devices shall be digital. The recorder or data acquisition system shall be capable of producing a secure and irrefutable batch record (cannot be altered without detection and cannot be a CSV file unless encrypted), specific to parts/items being heat treated, and include all reference information such as batch number, alloy, thickness, quantity, nomenclature, time, temperature, reference specification, and other information relevant. The recorder shall be set up as Instrumentation Type D as a minimum where the highest and lowest temperature locations of the established (or most recent) TUS are recorded as part of the batch record in addition to the chamber control temperature. A “Load Thermocouple” jack shall be provided to allow the configuration of Instrumentation Type A or B if needed or to verify heating characteristics of a particular part.

3.6.7 Excess temperature (high limit) controller. An excess temperature controller, independent of the temperature controller, shall be provided. The excess temperature controller shall automatically turn off the heating elements and sound an audible alarm when the chamber temperature rises above its set point temperature. Typically, the limit controller is set at the maximum operating temperature of the chamber, or at the limit of the class tolerance range to protect a load. All excess temperature controllers shall be equipped with temperature indication and a manual reset function, requiring the operator to acknowledge the excess temperature fault condition before the furnace is again operational. The temperature-sensing element of the excess temperature limit controller shall be suitable for the temperature and atmosphere to which they are exposed.

3.6.8 Quench tank. A quench tank shall be included for single chamber type I, range 2 and 3 furnaces for either water or oil quench respectively. For dual chamber type II where the chambers are range 1 and 2, a water quench tank shall be included.

3.6.8.1 Water quench tank. A water quench tank shall be sized to assure that the largest load to be quenched shall not increase the water temperature by more than 10 °F for aluminum sheet metal or 25 °F for castings.

3.6.8.2 Oil quench tank. An oil quench tank shall be sized to assure that the largest load to be quenched shall not increase the oil temperature by more than 40 °F. A type II furnace with a ranges 2 and 3 chambers shall include a dual bath quench, for both water and oil.

3.6.8.3 Quench tank construction. Construction of quench tanks shall be no less than 10 gauge stainless steel, and designed to manually roll under the furnace.

3.6.8.3.1 Water quench tank. Water quench tanks shall have a means of agitation via pump or impeller, or similar, or by mechanical means. Air agitation shall not be used.



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3.6.8.3.2 Oil quench tank. Oil quench tanks do not require agitation, unless specified by the contract.

3.6.9 Electrical system.

3.6.9.1 Design and construction. The electrical system design and construction shall conform to current National Fire Protection Association (NFPA) standards, NFPA 70, National Electrical Code and NFPA 79, Electrical Standard for Industrial Machinery. The design of the overall safety features and sequence of operation will be in accordance with NFPA 86, Standard for Ovens and Furnaces. The furnace power system shall be designed for operation with a 207 – 240 VAC or 440 – 480 VAC, 3 Phase, 60 Hz supply unless otherwise agreed to by the Government.

3.6.9.2 Accessories. The furnace shall be equipped with the following accessories: system power disconnect switch, properly sized fuses and/or circuit breakers, surge protection, indication lights, switches and pushbuttons in accordance with the requirements of the above specified standards.

3.6.10 Thermocouples. Thermocouples used for process control and recording shall meet all requirements of SAE AMS 2750. Each shall have an individual serial number with a “lot” certification for process thermocouples (all from the same spool that share certification), contain usage tags, be expendable by definition, and include sufficient sets/quantities for a 3 year supply with spares. Spare shall comply with CAGE 98752, Print 202113172, design.

3.6.10.1 Certification. All thermocouples shall have a certification. All certifications shall be performed by an accredited laboratory or supplier and be traceable to the National Institute of Standards and Technology (NIST).

3.7 Performance. The furnace shall be capable of operating continuously with the maximum rated load at any set temperature within the operating range considered by the manufacturer as standard without causing adverse effect to the furnace.

3.7.1 Exterior surface temperature. The furnace exterior allowable maximum surface temperature shall be no greater than 160 °F. If operating in excess of 160 °F, the furnace shall be guarded by location, shields, or insulation to prevent contact with personnel.

3.7.2 Recovery rate and heating capacity. The furnace shall have a recovery rate for not less than the weight of iron or steel per square foot of hearth area per hour heated to the temperature shown in table IV. The furnace shall have a heating capacity of not less than the weight of iron or steel per square foot of hearth area heated to the temperature shown in table V.



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TABLE IV. Recovery rate.

| Style | Range | Chamber temperature, Operation (°F) | Recovery rate, pounds/square foot/hour |
|---|---|---|---|
| A | 1 | 500 | 40 |
| | 2 | 1200 | 40 |
| B | 2 | 1200 | 60 |
| | 3 | 2000 | 60 |

TABLE V. Heating capacity.

| Style | Range | Chamber temperature, Operation | Heat capacity, pounds/square foot |
|---|---|---|---|
| A | 1 | 500 | 60 |
| | 2 | 1200 | 60 |
| B | 2 | 1200 | 100 |
| | 3 | 2000 | 100 |

3.8 Furnace dimensions and weight. If required, the maximum exterior dimensions (length, width, and height) and weight shall not exceed the restrictions specified per the contract purchase order.

3.9 Safety and health requirements. The furnace or oven shall be designed and manufactured in accordance with NFPA 86. If required to process volatile substances, combustible materials, or finished or oil-coated metal parts, a separate furnace or oven shall be acquired, which is specifically designed to meet NFPA 86 class A oven requirements. Otherwise, the furnace or oven shall meet NFPA 86, class B oven requirements. The manufacturer shall ensure that the furnace or oven and all equipment and accessories used on the furnace or oven shall be in compliance with Occupational Safety and Health Administration's (OSHA), Title 29 of the Code of Federal Regulations (CFR), Part 1910, Occupational Safety and Health Standards. If a conflict arises between the NFPA and OSHA standards, the OSHA standards shall apply. Warnings and identifications shall be visible and/or provisions shall be made to prevent or restrict entry into the oven where the atmosphere could be hazardous to human health.

3.10 System of units. The U.S. Customary System of Units (US) and the International System of Units (SI) shall be used to graduate measuring and indicating devices. Measuring and indicating devices on the furnace shall be graduated in the same system. Measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.11 Identification plate. An identification plate shall be securely attached to the furnace or oven. At minimum, the identification plate shall contain the following information:

a. Nomenclature.

b. Manufacturer's name/trademark.

c. Manufacturer’s serial number.

d. Furnace model designation.

e. Description of intended usage or applications

f. Power input (volts, phase, frequency, full load amperes (FLA), and required supply amperes).

g. Power rating of the heating elements (kW) and maximum ampere rating of the heating



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elements per chamber and both if dual chamber

h. Inert gas type, pressure and volume if included for range 3 furnaces

i. Design maximum temperature, and optimal temperature range.

j. Short-circuit/over-current rating.

k. Largest motor size, horsepower and nominal FLA).

l. Electrical diagram numbers

m. Contract number or order number.

n. National stock number.

o. Date of manufacture.

3.12 Lubrication plate or chart. A permanently-marked, lubrication plate or chart shall be attached to the furnace or oven. The information provided on the plate or chart shall include:

a. Points of lubricant application.

b. Servicing interval.

c. Type of lubricant(s) with SAE number or lubricant identifier.

3.13 Qualifying, training services, and instrument calibration. As part of commissioning, qualifying, and training services, the supplier may use Government owned equipment for the initial qualifying TUS and SAT provided that such equipment has a current calibration in compliance to SAE AMS 2750 and/or USAF Precision Measurement Equipment Laboratory (PMEL). Training is to include familiarization of the equipment, operating and safety procedures, resulting in OEM certified AF technicians for heat treating utilizing OEM equipment, and properly performing a TUS as well as completing compliant TUS and SAT reports.

3.14 Training for operators, pyrometry records management, maintenance, and trainers.

Immediately upon completion of oven certification per 5.3, and as required by SAE AMS 2750, the vendor shall conduct three days of operational training as follows:

a. Provide full training on fundamentals of heat treating for the various aviation grade metals and the type of furnace or oven provided, as well as all instructions for safe operation for charging the furnace or oven chamber(s), quenching, and general operation.

b. Provide training on how to operate all controls, heat treat various metals, set up and execute batch records, utilizing the monitoring and recording equipment.

c. Install any required software on a Government provided computer and establish communications between the furnace controls for data transfer and remote monitoring / control, as required.

d. Demonstrate and train all site personnel on how to perform a TUS and SAT along with records management. Training on the use of the Super Systems 8120 Portable Data Logger/TUS recorder is required, including verifying proper computer software installation, set up of templates, managing spool correction factors and other variables, and printing of TUS Reports.

e. Provide training on use of a Super Systems 8120 Portable Data Logger for performing required SAT reports presented with required details.

3.15 Warranty. The manufacturer's standard commercial warranty, terms, and conditions shall apply.

3.16 Requirements waiver. Requirements identified herein shall not be waived without the written



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consent of the Air Force Metals Technology Office/Metal Fabrication and Technologies Office, located in Warner Robins, GA.

3.17 Manuals. Additionally a complete wiring diagram, complete bill of materials, sequence of operation description, troubleshooting guide, and recommended spare parts shall be included as required by NFPA standards.

4. REGULATORY REQUIREMENTS.

4.1 Recovered, recycled, or biobased materials. The offeror/contractor is encouraged to use recovered, recycled, biobased materials to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403.

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance via inspection, testing and/or in writing. For all testing and inspections listed herein, the contactor shall provide test inspections and reports that detail findings and highlight compliance and/or non-compliance for each section.

5.2 Inspection. The furnace shall be inspected to determine compliance with all requirements specified in this CID after delivery to the intended Government facility. The contractor shall support the Government during this process (to include an inspection checklist creation and evaluation), if required.

5.2.1 Testing. Unless otherwise specified, the furnace or oven shall be tested in accordance with sections 3.1 through 3.8. Testing shall be witnessed and signed off (approved/rejected as compliant with requirements herein) by the Air Force Metals Technology Office/Metal Fabrication and Technologies Office, or their designated representatives located at the specified destination site/facility during commissioning services.

5.2.2 Test setup. The manufacturer shall be responsible for coordinating installation location(s), supplying all manpower, tooling and material(s) and shall conduct all tests required by the Government including the temperature uniformity survey test(s) with radiation panel test(s) and SAT. The manufacturer shall provide test instruments and equipment with calibration traceable to the NIST and shall provide certification of calibration. The furnace shall be tested at an ambient temperature of 80 °F, ±20 °F. Suppliers shall provide original equipment manufacturer (OEM) certificates of calibration for all instruments and recorder channels or they may be provided by a third-party independent service company.

5.2.3 Operational test. The furnace or oven shall be operated in accordance with the manufacturer's standard operational procedures for furnace startup and refractory cure. The furnace shall be operated at the chamber maximum rated temperature for not less than 8 hours continuous. Proper operation of all controls and features shall be verified, documented, and certified by the manufacturer during the trial period. Operational test procedures shall include sample parts being heat treated with step by step procedures of how to operate the furnace. Comprehensive documentation of operational test shall be provided to the Government.



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5.2.4 Surface temperature test. The chamber shall be stabilized at its maximum operating temperature for one hour (minimum), and then the exterior surface temperatures of the furnace and door shall be measured at random locations (at least 10 locations, each side of the oven, separated by nine-inches or more). The exterior surfaces shall not exceed 160 °F.

5.2.5 Performance test. The furnace, with the chamber temperature stabilized at the temperature shown in table IV, shall be loaded with an ambient-temperature batch of aluminum, iron, or steel equivalent to the hourly recovery weight designated for the furnace style and range. A temperature survey shall be recorded in accordance with SAE AMS 2750. This is typically a nine point survey consisting of the eight corners of the defined work volume and the ninth in the center. For work volumes of three cubic feet or less, a five point survey is allowed in accordance with SAE AMS 2750. The TUS may be performed with probes or survey wire consistent with usage restrictions for nonexpendable or expendable thermocouples. During the recovery cycle, and at the end of one hour, the temperature shall have recovered within ±10 °F for style A, and ±25 °F for style B furnaces of the initial temperature. From the time the recovery cycle is initiated until thermal equilibrium is reached, no temperature reading shall exceed the target recovery temperature by more than ±10 °F for style A, and ±25 °F for style B furnaces. The results of the test shall be compared for compliance with the requirements specified in section 3.7. Documentation of the performance test shall be provided to the Government.

5.2.6 Temperature uniformity test (TUS). After the load and chamber in section 5.2.5 performance test have reached thermal equilibrium, a temperature survey shall be taken every two minutes (maximum) for 30 minutes in accordance with SAE AMS 2750. The temperature at the control thermocouple shall not vary more than ±2 °F for style A or B furnaces or ovens from the set temperature after recovery and stabilization. The temperature variation between the chamber test points and the control thermocouple shall not exceed ±10 °F for style A, and ±25 °F style B furnaces or ovens. Alternatively, as permitted by SAE AMS 2750 and after the performance test in 5.2.5 is complete, the TUS may be performed with an open chamber (no load) after reaching thermal equilibrium. The furnace door is opened briefly simulating placement of a load in the chamber and allowing the indicated temperature to drop at least 100 °F, then after closing the door the furnace shall not overshoot with any of the typically nine survey point locations by 10° F for style A, and by ±25 °F style B furnaces. How the initial TUS is performed will become the established method of the required periodic TUS procedure.

5.2.6.1 Test documentation. Documentation of temperature uniformity shall be provided to the Government.

5.2.6.2 Qualifying document/TUS charts. The furnace or oven system shall have a chart for each chamber that illustrates the location of the TUS survey points and dimensions of the TUS volume. An additional document will state the results of the initial qualifying TUS, including the maximum temperature for the TUS performed, position and readings of TUS high/low. The document will state the instrumentation type (A, B, C or D) and if qualified as class 1, 2, or 3. Additionally the document will list SAE AMS 2750 as the referenced standard for compliance and qualifying. Both documents shall be displayed prominently on the side of the furnace system.

5.2.6.3 Optional tests. Optional tests shall be conducted as determined by the Air Force Metals Technology Office/Metal Fabrication and Technologies Office.

5.3 Acceptance criteria. Preliminary and final testing as specified in 5.2.1 through 5.2.6.3 shall be conducted at the Government receiving site, as directed by the Air Force Metals Technology Office/Metal Fabrication and Technologies Office. The furnace system meets all qualifying criteria and be certified for heat treating aircraft parts, with full training prior to acceptance. Failure of the



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furnace and/or required acceptance services to meet these requirements shall be cause for rejection.

6. PACKAGING.

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order.

7. NOTES.

7.1 Source of documents.

7.1.1 Government documents. Electronic copies of specifications, standards, and handbooks are available at https://assist.dla.mil/.

7.1.2 Code of Federal Regulations (CFR). Electronic copies of the eCFRs are available at https://www.ecfr.gov/.

7.1.3 Environmental Protection Agency (EPA). Electronic copies of these documents are available at https://www.epa.gov/.

7.1.4 Federal Acquisition Regulations (FAR). Electronic copies of the FAR are available at https://www.acquisition.gov/.

7.1.5 National Institute of Standards and Technology (NIST) documents. Electronic copies of NIST documents are available at https://www.nist.gov.

7.1.6 Occupational Safety and Health Association (OSHA). Electronic copies of OSHA documents are available at https://www.osha.gov/.

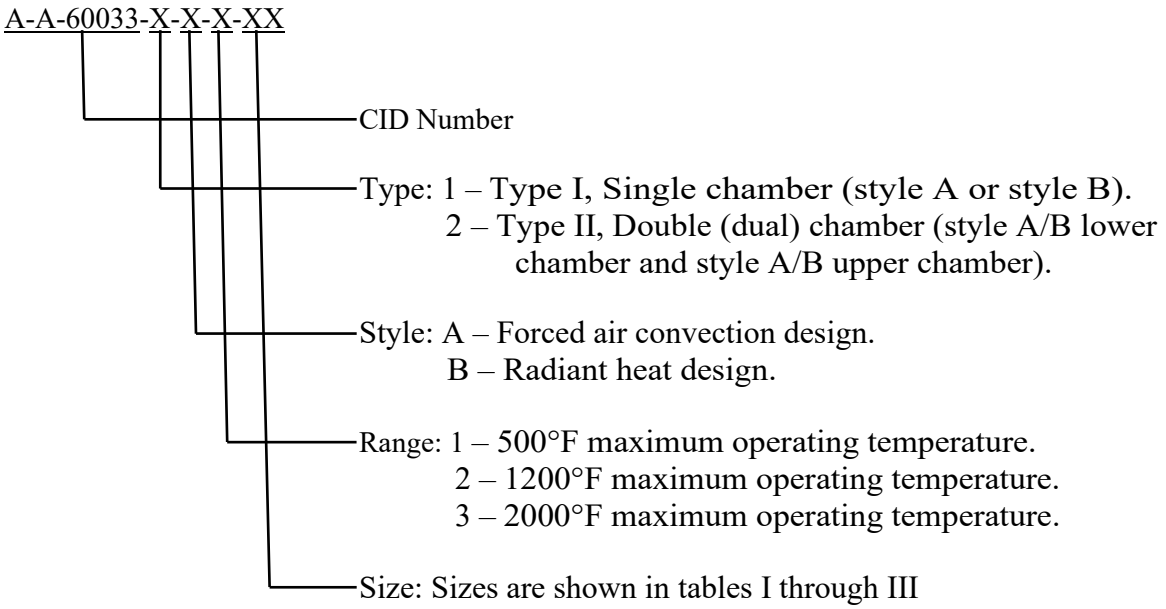
7.1.7 NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) documents.
(Copies of NFPA standards are available online at https://www.nfpa.org/.)

7.1.8 SOCIETY OF AUTOMOTIVE ENGINEERS (SAE).
(Copies of SAE standards are available online at https://www.sae.org/.)



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7.2 Part or identification number (PIN). The following part or identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.




7.3 Key Words.
Furnace, Oven,
Heat Treat,
Heat Treating,
Temperature Uniformity Tests,
System Accuracy Tests

MILITARY INTERESTS:

Custodian:
Air Force – 184

Reviewer:
DLA – GS

Preparing activity:
Air Force – 184

Agent activity:
Air Force – 110

(Project: 3424-2023-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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